

Arm

Inverse Kinematics:

2 DOF, 3 DOF,

$$\cos(\theta) = (x^2 + y^2 - L_1^2 - L_2^2) / (2L_1 L_2)$$

6 DOF:

Equation: $W_c = P - d_6 \cdot R_z$

Where:

- **W_c:** Wrist Centre
- **P:** Desired gripper position
- **d₆:** Length of gripper
- **R_z:** Orientation described by rotation matrix

Dimensions of arm:

Let the arm be attached at **45 cm** of height from base.

Spearhead Rack

Rod Rack

Height of spear to be picked up = 50cm + 5.25cm = 55.25cm

Link1 = 10cm, Link2 = 22.5cm, Link3 = 22.5cm, Link4 = 5cm,

Link5 = 5cm, Link6 = 5cm

Total = 60cm arm length

Reach Calculations:

- Lowest height of reach = **55.25 cm**
- Highest height of reach = **100 cm**

[Considering 700³cm³ volume.]

Shoulder torque:

Equation: $\tau = (L_{arm} \times W_{obj}) + (L_{cg} \times W_{arm})$

Where:

- τ : Torque
- L_{arm} : Length of arm
- W_{obj} : Weight of object
- L_{cg} : Length to Center of Gravity (CG)
- W_{arm} : Weight of arm

Factor of Safety (FOS):

- Load capacity considered = 500 gram

Material:

- Arm made of aluminium / plastic

Weight Distribution (Total arm weight = 1 kg):

- Link 1 — 250 g
- Link 2 — 300 g
- Link 3 — 250 g
- Gripper — 200 g

J3 Calculation (Forearm, wrist & payload):

Assumption: Max moment is calculated, thinking arm is horizontal.

Equation: $\tau_{J3} = [m_3 \times (l_3 / 2)] + (m_w \times l_3) + (m_p \times [l_3 + l_w])$

Substitution: $\tau_{J3} = [0.25 \times 11.25] + (0.2 \times 22.5) + (0.5 \times 27.5)$

Result: $\tau_{J3} = 21.06 \text{ kg-cm}$

J2 Calculation:

Equation: $\tau_{J2} = M_{L2} + M_{L3} + M_w + M_{payload}$

Result: $\tau_{J2} = 45.81 \text{ kg-cm}$

Dynamic Torque:

Formula: $T_{\text{Total}} = T_{\text{Static}} + \text{variable}$

Calculations:

- $J_2: 45.81 \times 2 = 91 \text{ kg-cm}$
- $J_3: 21.06 \times 2 = 42 \text{ kg-cm}$
- $J_4 - J_6: 15 \text{ kg-cm (Wrist)}$

For base,

Parameters:

- $\alpha = 5.23 \text{ rad/s}^2$ (approx)
- $m = 1.5 \text{ kg}$
- $\text{max reach} = 50 \text{ cm}$

Torque Calculations:

- $T_{\text{inertia}} = 0.653 \text{ N}\cdot\text{m}$
- $T_{\text{frictional}} = 0.15 \text{ N}\cdot\text{m}$

Base Torque:

- $T_{\text{base}} = 8.19 \text{ kg-cm} \times 3$

Final Result: $\tau = 24.57 \text{ kg-cm}$

